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BIO 101: Introduction to Biology - Cells, Genetics and Diversity

The organising ideas of life, from the cell to ecosystems

FACULTY PRIMER

Faculty Primer

Faculty of Bio-Sciences
Botany | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

BIO 101 covers an enormous syllabus, so the danger is memorising isolated facts. This pack organises the course around four unifying ideas - cells, information, energy and evolution - and gives you diagram-and-label practice for the questions that always appear.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Compare prokaryotic and eukaryotic cells and describe the functions of major organelles.
- Explain the cell cycle, mitosis and meiosis and why the distinction matters.
- Apply Mendelian genetics to simple crosses and interpret pedigree information.
- Classify organisms using binomial nomenclature and explain the basis of biological diversity.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

Cells as the unit of life

Every living thing is made of cells, and every cell maintains a boundary, processes energy and stores information.

- Prokaryotes lack a membrane-bound nucleus; eukaryotes have one.
- Nucleus: stores DNA and coordinates gene expression.
- Mitochondria: site of aerobic respiration and ATP production.
- Chloroplasts: photosynthesis in plants and algae.
- Ribosomes: protein synthesis; rough endoplasmic reticulum folds proteins.

- Cell membrane: selectively permeable phospholipid bilayer controlling exchange.

> **Label diagrams in the exam**
> Draw large, label with straight lines, and give a function beside each label. Marks are usually split between identification and function.

Cell division and inheritance

Mitosis produces identical cells for growth; meiosis produces gametes with half the chromosome number.

- Mitosis: one division, two diploid daughter cells, genetically identical.
- Meiosis: two divisions, four haploid cells, genetic variation through crossing over and independent assortment.
- Genotype is the genetic makeup; phenotype is the observable trait.
- A dominant allele is expressed when present; a recessive allele is expressed only when homozygous.
- Use a Punnett square for monohybrid crosses and state ratios clearly.
- Pedigrees can indicate whether a trait is dominant, recessive, autosomal or sex-linked.

Energy, ecology and diversity

Life persists by capturing energy and cycling matter through ecosystems.

- Photosynthesis converts light energy into chemical energy; respiration releases it.
- Food chains show energy flow; only about 10 percent passes to the next trophic level.
- Nutrient cycles such as carbon, nitrogen and water connect living and non-living systems.
- Binomial nomenclature: genus capitalised, species lowercase, both italicised.
- Classification hierarchy: domain, kingdom, phylum, class, order, family, genus, species.
- Variation plus selection is the mechanism behind adaptation and speciation.

The levels of organisation

Biology studies living systems at every scale: molecules, organelles, cells, tissues, organs, organisms, populations, communities and ecosystems. The same question - how is this organised and how does it maintain itself - gets a different answer at each level, and part of learning the subject is knowing which level a question belongs to.

The unifying ideas are few and powerful: cells are the basic unit of life; all living things use the same molecular machinery (DNA, RNA, proteins, ATP); structure and function are related at every level; and evolution by natural selection explains both diversity and similarity.

A cell is not a bag. It is compartmentalised: the membrane regulates exchange, the nucleus holds genetic material in eukaryotes, mitochondria generate ATP, ribosomes make protein, and the endomembrane system moves material. Prokaryotes lack membrane-bound organelles and are nonetheless biochemically sophisticated.

Information, energy and inheritance

DNA stores information in a four-letter code; transcription produces RNA; translation produces protein. The genetic code is nearly universal, which is itself evidence of common ancestry. Mutations alter the sequence, and their consequences range from silent to lethal.

Energy flow follows thermodynamics. Photosynthesis captures light energy and converts it into chemical bonds; cellular respiration releases that energy as ATP. ATP is the currency, not the store. Enzymes lower activation energy without changing the equilibrium, and their activity depends on temperature, pH and substrate concentration.

Inheritance follows Mendelian ratios for simple traits, with complications that are now standard: incomplete dominance, codominance, multiple alleles, linked genes, polygenic traits and environmental effects. Most real traits are not one-gene-one-trait.

> **Structure predicts function**
> For any biological structure you are asked to describe, ask what the shape allows and what it prevents. This single habit answers most "explain the significance" questions.

Ecology, evolution and the Nigerian context

Evolution by natural selection requires variation, heritability and differential reproductive success. It is both a historical account of life's diversification and a mechanism operating in the present, visible in antibiotic resistance and in crop pest adaptation.

Ecology studies interactions between organisms and environment: competition, predation, parasitism, mutualism and succession. Nutrient cycling and energy flow explain why ecosystems have limits. Human activity - habitat loss, pollution, overexploitation, climate change - is now the dominant driver of biodiversity loss.

Nigeria's biological context is distinctive: rainforest and savanna, high endemism in some taxa, and severe pressures from deforestation, wetlands loss and urban expansion. Local examples make exam answers stronger than generic ones.

WORKED EXAMPLES

Do it once, slowly

These are not model answers to copy. They show the reasoning a strong student would use, step by step, so you can repeat the process on a question you have not seen.

1. Following a genetic cross

Two heterozygous tall pea plants (Tt) are crossed. Predict the offspring.

1. Identify the alleles: T dominant (tall), t recessive (short).
2. Write the gametes each parent can produce: T and t.
3. Construct a Punnett square: TT, Tt, Tt, tt.
4. State the genotypic ratio: 1 TT : 2 Tt : 1 tt.
5. State the phenotypic ratio: 3 tall : 1 short.
6. Note the assumption: independent assortment and no lethality.

> **What to carry forward**
> Gametes first, square second, both ratios third. State assumptions.

2. Explaining an experimental result

An enzyme's rate rises with temperature to 40 degrees then falls sharply. Explain.

1. State the relationship: kinetic energy increases collisions up to an optimum.
2. Explain the rise: more frequent effective collisions between enzyme and substrate.
3. Explain the fall: denaturation of the active site above the optimum.
4. Distinguish reversible slowing from irreversible denaturation.
5. Note that the optimum is specific to the enzyme and its environment.

> **What to carry forward**
> Two mechanisms, opposite directions. Name both and where the switch occurs.

EXAM TECHNIQUE

Questions with model answers

Written in the shape your course is usually examined, with the marking logic spelled out. Attempt each one under time before you read the outline.

Q1. Explain how the structure of the cell membrane relates to its functions.

15 marks

1. Describe the phospholipid bilayer and embedded proteins.
2. Explain selective permeability.
3. Distinguish diffusion, facilitated diffusion and active transport.
4. Link each function to a structural feature.

Q2. Discuss the evidence for evolution by natural selection.

20 marks

1. State the mechanism and its three requirements.
2. Fossil evidence and transitional forms.
3. Comparative anatomy and homologous structures.
4. Molecular evidence: universal code, sequence similarity.
5. Observed contemporary evolution: resistance.
6. Address one objection and its reply.

REFERENCE

Glossary

Use these definitions in your own sentences; a copied definition rarely earns full marks.

Eukaryote. An organism whose cells contain membrane-bound organelles and a nucleus.

Prokaryote. An organism lacking a membrane-bound nucleus and organelles.

ATP. Adenosine triphosphate, the cell's energy-transfer molecule.

Enzyme. A protein catalyst that lowers activation energy for a reaction.

Denaturation. Loss of a protein's functional shape through heat, pH or chemical action.

Allele. One of the variant forms of a gene.

Genotype. The genetic makeup of an organism at a locus.

Phenotype. The observable characteristics produced by genotype and environment.

Natural selection. Differential reproductive success of heritable variants.

Homeostasis. Maintenance of a stable internal environment despite external change.

Trophic level. A feeding position in a food chain.

Endemism. The condition of a species being native to a single region and found nowhere else.

Your four-week plan

Week 1 - Cells

Structure and function.

- Draw and label a eukaryotic cell.
- Match five organelles to functions.
- Compare prokaryote and eukaryote.

Week 2 - Biochemistry

Energy and enzymes.

- Summarise photosynthesis and respiration.
- Explain enzyme kinetics with a graph.
- State two factors affecting enzyme activity.

Week 3 - Genetics

Inheritance.

- Solve five monohybrid crosses.
- Explain one non-Mendelian case.
- Describe a mutation and its effect.

Week 4 - Ecology

Systems.

- Draw a food web from a local habitat.
- Explain one nutrient cycle.
- Describe one local conservation pressure.

Where to read more

Entry-level texts you can find in the university library or a departmental reading list. Confirm editions with your lecturer.

- Any first-year biology textbook covering cell biology, genetics and ecology.
- A genetics chapter with worked monohybrid and dihybrid problems.
- A Nigerian ecology or conservation source for local examples.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. State one difference between mitosis and meiosis.

Answer: Mitosis produces two genetically identical diploid cells; meiosis produces four genetically varied haploid cells.

2. Why are mitochondria called the powerhouse of the cell?

Answer: They carry out aerobic respiration, producing most of the cell's ATP, the energy currency.

3. Cross Aa with Aa. State the expected genotype ratio.

Answer: AA : Aa : aa = 1 : 2 : 1, giving a 3:1 phenotypic ratio if A is completely dominant.

4. Write the correct binomial format for the human species.

Answer: Homo sapiens - genus capitalised, species lowercase, both italicised.

5. Why is energy lost between trophic levels?

Answer: Organisms use much of their energy for respiration, movement, growth and waste, so only a fraction is stored as biomass available to the next level.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Draw and label a eukaryotic cell with functions for eight organelles.
- [ ] Make a comparison table of mitosis and meiosis.
- [ ] Solve five monohybrid crosses and one test cross.
- [ ] Learn the classification hierarchy and place five organisms.
- [ ] Draw a food web with at least six organisms and label trophic levels.
- [ ] Define ten biology terms precisely without examples.
- [ ] Answer one past BIO 101 question with diagrams under timed conditions.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date:

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